A correction to "An arithmetic regularity lemma, an associated counting lemma, and applications"

Terence Tao · 26 November 2020

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Ben Green and I have updated our paper "An arithmetic regularity lemma, an associated counting lemma, and applications" to account for a somewhat serious issue with the paper that was pointed out to us recently by Daniel Altman. This paper contains two core theorems:

- An "arithmetic regularity lemma" that, roughly speaking, decomposes an arbitrary bounded sequence $f(n)$ on an interval $\{1,\dots,N\}$ as an "irrational nilsequence" $F(g(n)\Gamma)$ of controlled complexity, plus some "negligible" errors (where one uses the Gowers uniformity norm as the main norm to control the neglibility of the error); and
- An "arithmetic counting lemma" that gives an asymptotic formula for counting various averages $\mathbb{E}_{\mathbf{n}\in\mathbf{Z}^d\cap P} f(\psi_1(\mathbf{n}))\dots f(\psi_t(\mathbf{n}))$ for various affine-linear forms $\psi_1,\dots,\psi_t$ when the functions f are given by irrational nilsequences.

The combination of the two theorems is then used to address various questions in additive combinatorics.

There are no direct issues with the arithmetic regularity lemma. However, it turns out that the arithmetic counting lemma is only true if one imposes an additional property (which we call the "flag property") on the affine-linear forms $\psi_1,\dots,\psi_t$. Without this property, there does not appear to be a clean asymptotic formula for these averages if the only hypothesis one places on the underlying nilsequences is irrationality. Thus when trying to understand the asymptotics of averages involving linear forms that do not obey the flag property, the paradigm of understanding these averages via a combination of the regularity lemma and a counting lemma seems to require some significant revision (in particular, one would probably have to replace the existing regularity lemma with some variant, despite the fact that the lemma is still technically true in this setting). Fortunately, for most applications studied to date (including the important subclass of translation-invariant affine forms), the flag property holds; however our claim in the paper to have resolved a conjecture of Gowers and Wolf on the true complexity of systems of affine forms must now be narrowed, as our methods only verify this conjecture under the assumption of the flag property.

In a bit more detail: the asymptotic formula for our counting lemma involved some finite-dimensional vector spaces $\Psi^{[i]}$ for various natural numbers i, defined as the linear span of the vectors $(\psi_1^i(\mathbf{n}),\dots,\psi_t^i(\mathbf{n}))$ as $\mathbf{n}$ ranges over the parameter space $\mathbf{Z}^d$. Roughly speaking, these spaces encode some constraints one would expect to see amongst the forms $\psi_1^i(\mathbf{n}),\dots,\psi_t^i(\mathbf{n})$. For instance, in the case of length four arithmetic progressions when $d=2$, $\mathbf{n}=(n,r)$, and

$$\psi_i(\mathbf{n}) = n + (i-1)r$$

for $i=1,2,3,4$, then $\Psi^{[1]}$ is spanned by the vectors $(1,1,1,1)$ and $(1,2,3,4)$ and can thus be described as the two-dimensional linear space

$$\Psi^{[1]} = \{(a,b,c,d) : a-2b+c = b-2c+d = 0\} \tag{1}$$

while $\Psi^{[2]}$ is spanned by the vectors $(1,1,1,1)$, $(1,2,3,4)$, $(1^2,2^2,3^2,4^2)$ and can be described as the hyperplane

$$\Psi^{[2]} = \{(a,b,c,d) : a - 3b + 3c - d = 0\}. \tag{2}$$

As a special case of the counting lemma, we can check that if f takes the form $f(n) = F(\alpha n, \beta n^2 + \gamma n)$ for some irrational $\alpha, \beta \in \mathbf{R}/\mathbf{Z}$, some arbitrary $\gamma \in \mathbf{R}/\mathbf{Z}$, and some smooth $F : \mathbf{R}/\mathbf{Z} \times \mathbf{R}/\mathbf{Z} \to \mathbf{C}$, then the limiting value of the average

$$\mathbf{E}_{n,r\in[N]} f(n)f(n+r)f(n+2r)f(n+3r)$$

as $N \to \infty$ is equal to

$$\int_{a_1,b_1,c_1,d_1 \in \mathbf{R}/\mathbf{Z}: a_1 - 2b_1 + c_1 = b_1 - 2c_1 + d_1 = 0} \int_{a_2,b_2,c_2,d_2 \in \mathbf{R}/\mathbf{Z}: a_2 - 3b_2 + 3c_2 - d_2 = 0}$$
$$F(a_1,a_2)F(b_1,b_2)F(c_1,c_2)F(d_1,d_2)$$

which reflects the constraints

$$\alpha n - 2\alpha(n+r) + \alpha(n+2r) = \alpha(n+r) - 2\alpha(n+2r) + \alpha(n+3r) = 0$$

and

$$(\beta n^2 + \gamma n) - 3(\beta(n+r)^2 + \gamma(n+r))$$
$$+3(\beta(n+2r)^2 + \gamma(n+2r)) - (\beta(n+3r)^2 + \gamma(n+3r)) = 0.$$

These constraints follow from the descriptions (1), (2), using the containment $\Psi^{[1]} \subset \Psi^{[2]}$ to dispense with the lower order term γn (which then plays no further role in the analysis).

The arguments in our paper turn out to be perfectly correct under the assumption of the "flag property" that $\Psi^{[i]} \subset \Psi^{[i+1]}$ for all i. The problem is that the flag property turns out to not always hold. A counterexample, provided by Daniel Altman, involves the four linear forms

$$\psi_1(n,r) = r; \psi_2(n,r) = 2n + 2r; \psi_3(n,r) = n + 3r; \psi_4(n,r) = n.$$

Here it turns out that

$$\Psi^{[1]} = \{(a,b,c,d) : d - c = 3a; b - 2a = 2d\}$$

and

$$\Psi^{[2]} = \{(a,b,c,d) : 24a + 3b - 4c - 8d = 0\}$$

and $\Psi^{[1]}$ is no longer contained in $\Psi^{[2]}$. The analogue of the asymptotic formula given previously for $f(n) = F(\alpha n, \beta n^2 + \gamma n)$ is then valid when γ vanishes, but not when γ is non-zero, because the identity

$$24(\beta\psi_1(n,r)^2 + \gamma\psi_1(n,r)) + 3(\beta\psi_2(n,r)^2 + \gamma\psi_2(n,r))$$
$$-4(\beta\psi_3(n,r)^2 + \gamma\psi_3(n,r)) - 8(\beta\psi_4(n,r)^2 + \gamma\psi_4(n,r)) = 0$$

holds in the former case but not the latter. Thus the output of any purported arithmetic regularity lemma in this case is now sensitive to the lower order terms of the nilsequence and cannot be described in a uniform fashion for all "irrational" sequences. There should still be some sort of formula for the asymptotics from the general equidistribution theory of nilsequences, but it could be considerably more complicated than what is presented in this paper.

Fortunately, the flag property does hold in several key cases, most notably the translation invariant case when $\Psi^{[1]}$ contains $(1,...,1)$, as well as "complexity one" cases. Nevertheless non-flag property systems of affine forms do exist, thus limiting the range of applicability of the techniques in this paper. In particular, the conjecture of Gowers and Wolf (Theorem 1.13 in the paper) is now open again in the non-flag property case.

Comments (5)

 Q · 26 November 2020

No relation to main theme of posting. Eq 1 and 2 seems related to (x-y)^2 and (x-y)^3. What is the terminology for this combinatorics and does it change in a nice formulaic manner if the second and third vectors do not start at 1?

 **Anonymous** · 26 November 2020

Is it possible to partition the lower order terms into several classes (with several flag properties) where each class obeys its particular flag-property?

 **Anonymous** · 27 November 2020

I think the first n in the first display should probably be $\mathbf{n}$
[Corrected, thanks – T.]

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[…] ; Ben Green and I thought we had resolved this conjecture back in 2010, though it turned out there was a subtle gap in our arguments and we were only able to resolve the conjecture in a partial range of cases. However, the full […]